

Problem E. String er khela

Time limit 2000 ms

Mem limit 262144 kB

Given a string x of length n and a string s of length m ($n \cdot m \leq 25$), consisting of lowercase Latin letters, you can apply any number of operations to the string x .

In one operation, you append the current value of x to the end of the string x . Note that the value of x will change after this.

For example, if $x = \text{"aba"}$, then after applying operations, x will change as follows: $\text{"aba"} \rightarrow \text{"abaaba"} \rightarrow \text{"abaabaabaaba"}$.

After what **minimum** number of operations s will appear in x as a substring? A substring of a string is defined as a **contiguous** segment of it.

Input

The first line of the input contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains two numbers n and m ($1 \leq n \cdot m \leq 25$) — the lengths of strings x and s , respectively.

The second line of each test case contains the string x of length n .

The third line of each test case contains the string s of length m .

Output

For each test case, output a single number — the **minimum** number of operations after which s will appear in x as a substring. If this is not possible, output -1 .

Examples

Input	Output
12	3
1 5	1
a	2
aaaaa	-1
5 5	1
eforc	0
force	1
2 5	3
ab	1
ababa	0
3 5	2
aba	5
ababa	
4 3	
babb	
bbb	
5 1	
aaaaa	
a	
4 2	
aabb	
ba	
2 8	
bk	
kbbkbbkb	
12 2	
fjdgmujlcont	
tf	
2 2	
aa	
aa	
3 5	
abb	
babba	
1 19	
m	
oooooooooooooooooooo	

Note

In the first test case of the example, after 2 operations, the string will become "aaaa", and after 3 operations, it will become "aaaaaaaa", so the answer is 3.

In the second test case of the example, after applying 1 operation, the string will become "e**forceforc**", where the substring is highlighted in red.

In the fourth test case of the example, it can be shown that it is impossible to obtain the desired string as a substring.